

Distributed Machine Learning Course Syllabus:

Introduction to Parallel and Distributed Models of Computation (~2 Lectures)

Week 1. Foundations of Parallel and Distributed Computing

- *Basics of Parallel Computing: Definitions, Importance, and Applications*
- *Introduction to Distributed Systems: Characteristics and Design Goals*
- *Scalable Frameworks for ML: An Overview*
- *Understanding Data and Computation Heterogeneity*

Week 2. Data Parallelism vs. Model Parallelism

- *Definitions and Key Differences*
- *Advantages and Limitations of Each Approach*
- *Case Studies: Real-World Applications and Decision Factors*

Challenges in Distributed ML (~2 Lectures)

Week 3. Challenges in Distributed ML

- *Consistency Models and Techniques*
- *Fault Tolerance Mechanisms in Distributed Systems*
- *Communication Overheads: Issues and Optimization Techniques*
- *Resource Management Challenges*

Week 4. Solutions and Strategies

- *Advanced Programming Models for Distributed ML*
- *Techniques for Efficient Resource Allocation and Management*
- *Strategies for Enhancing Fault Tolerance*
- *Communication Efficiency Improvements*

Algorithm Scaling Examples (~8 Lectures)

Week 5. Distributed ML Algorithms

- *K-means and DBSCAN: Distributed Implementations and Use Cases*
- *Distributed Association Rule Mining: Focus on FDM*
- *Distributed algorithm for SVM and random forest*

Week 6: Linear and Logistic Regression in Distributed Environments

- *Adaptation of Regression Models for Distributed Computing*
- *Performance Analysis and Optimization Techniques*

Week 7: Distributed Deep Learning Algorithms - Part 1

- *Gradient Descent Techniques for Empirical Risk Minimization*
- *SGD in Neural Network Training: Theory, Application, and Convergence Analysis*

Week 8: Distributed Deep Learning Algorithms - Part 2

- *Distributed Synchronous SGD: Concepts and Challenges*
- *Asynchronous SGD: Approaches, Benefits, and Drawbacks*

Week 9: Advanced SGD Techniques

- *Decentralized SGD: Methodology and Applications*
- *Overlap and Quantized SGD: Innovations and Impacts*

Week 10: Adacomm Vs Distributed SGD

- *Adaptive Communication Strategies: Theory and Comparison*
- *Case Studies on Efficiency and Scalability*

Week 11: Innovations in SGD

- *Elastic Averaging, AdaSyn, AdaQuant: Contributions to Distributed DL*
- *Practical Implications and Performance Evaluations*

Week 12: Federated Learning

- *Privacy and Security Considerations*
- *Hyperparameter Optimization in a Distributed Setting*

Week 13-14: Example Case Studies from Industry (~4 Lectures)

Week 13: Introduction to Industry Case Studies

Objective: Set the stage for in-depth industry case studies by highlighting the importance of distributed ML in solving real-world problems. Discuss criteria for selecting case studies and methodologies for analysis.

Week 13: Case Study 1 - Healthcare: Predictive Analytics for Patient Care

Organization: A leading healthcare provider or research institution.

Challenge: Managing and analyzing vast amounts of patient data from disparate sources to predict health outcomes and personalize patient care.

Solution: Implementation of a distributed ML framework to efficiently process and analyze data, utilizing algorithms like distributed logistic regression and federated learning to ensure data privacy and scalability.

Outcome: Improved accuracy in predictive analytics for patient care, leading to better health outcomes and optimized resource allocation.

Week 13: Case Study 2 - Financial Services: Fraud Detection at Scale

Organization: A global financial institution or fintech company.

Challenge: Detecting fraudulent transactions in real-time across millions of transactions globally, requiring high scalability and low latency.

Solution: Deployment of a distributed ML system using techniques such as distributed data parallelism and model parallelism for real-time fraud detection. Integration of asynchronous SGD for rapid model updates based on new fraud patterns.

Outcome: Significant reduction in fraudulent transactions, improved customer trust, and reduced operational costs.

Week 14: Case Study 3 - Retail and E-commerce: Personalized Recommendations at Scale

Organization: A major e-commerce platform.

Challenge: Delivering personalized product recommendations to millions of users in real-time, processing vast amounts of user interaction data.

Solution: Utilization of distributed deep learning algorithms, including decentralized SGD and federated learning, to train recommendation models on distributed datasets while preserving user privacy.

Outcome: Enhanced customer experience through personalized recommendations, increased sales, and customer retention.

Week 14: Case Study 4 - Autonomous Vehicles: Real-time Data Processing and Decision Making

Organization: A leading autonomous vehicle company.

Challenge: Processing and analyzing real-time data from various sensors and cameras for immediate decision-making, requiring low latency and high reliability.

Solution: Implementation of edge computing with distributed ML models, using techniques like local-update SGD and quantized SGD for efficient, real-time data processing and decision-making.

Outcome: Improved safety and reliability of autonomous vehicles, enabling more advanced functionalities and paving the way for broader adoption.

Evaluation:

Midterm Project (20%): Implementation and presentation of a distributed machine learning model.

Homework Assignments (30%): Weekly or bi-weekly assignments covering theoretical concepts and practical exercises.

Final Project (30%): Comprehensive project requiring students to design, implement, and evaluate a distributed ML solution for a real-world problem.

Participation and Quizzes (20%): Active participation in discussions and periodic quizzes on the course material.

References:

1. Book: Distributed Machine Learning with Python Accelerating model training and serving with distributed systems by Guanhua Wang, packet publishers.

2. Book: Distributed MachineLearning Patterns by YUAN TANG, Manning publishers.
3. Book: Federated Learning with Python, Kiyoshi Nakayama, PhD, Packet publishers.